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CHAPTER 0

What Is a Statistical Model?

In this chapter you will learn to:

- Discuss statistical models using the correct terminology.
- Employ a four-step process for statistical modeling.

This book is unified by a single theme, statistical models for data analysis. Such models can help to answer all kinds of questions, for example:

- Used cars. Can miles on the odometer predict the price? If so, how does each 1000 miles affect the price? What about the car's age as a predictor? If you know the mileage, does age matter? What about make and model? Do age and mileage give different price predictions for a BMW and a Maxima?
- Walking babies. Can special exercises help young babies walk earlier?
- Hawks' tails. Can we tell the species of a hawk from the length of its tail?
- Medical school admission. If your GPA is 3.5 and your friend's GPA is 3.6, how does that difference affect each of your chances of getting into medical school? What if one of you is male and the other female? What if you take both sex and GPA into account?
- Migraines. Can magnetic pulses to your head reduce your pain from a migraine?

- Ice cream. When people help themselves, do they take more if they have a bigger bowl? What about a bigger spoon?
- Golfers. Golf, like baseball, is a game for patient data nerds. Both games are slow, and both offer lots of things to measure. For example, in golf we measure driving distance, driving accuracy, putting performance, iron play, etc. In the end, what matters most is average strokes per hole. Which predictor or set of predictors works "best" for modeling overall score?
- Burgers. A national chain cooks their hamburgers 24 at a time in a 6 × 4 array. Health regulations require that each burger reach a fixed minimum temperature before you can sell it. The grill heats unevenly, and you don't want to overcook the burgers that heat faster. What should you measure?

Models can serve a variety of goals. Here are just seven, based on the preceding examples.

a. **Prediction:** Predict the price of a used car or the chance of getting into medical school.

b. **Classification:** Tell the species of a hawk based on its tail length.

c. **Evaluating a treatment:** Can special exercises help a baby walk sooner? Can magnetic pulses relieve migraine pain?

d. **Testing a theory:** Theory says you take more ice cream if you have a bigger bowl or a bigger spoon. What does the evidence say?

e. **Summarizing a pattern:** How are strokes per hole in golf related to distance and accuracy for drives, distance and accuracy for iron play, and distance and accuracy for putts?

f. **Improving a process:** What variables have an effect on how quickly a burger heats up?

g. **Making a decision:** The same burger chain needs to decide: Should we make all our burgers at a central location and ship them frozen, or should we make them fresh on site at each franchise?

CAUTION

Before we go into some details about models, we offer an important caution: *Every model simplifies reality.* No one would mistake a model airplane for the real thing. No one should make that same mistake with a statistical model. Just because the model comes dressed up with equations is no reason you have to trust it. Insist that the model prove its value. As one of the most influential statisticians of the last century (G. E. P. Box) said, "All models are wrong; some are useful." Remember Box. Always ask two questions,

"How far wrong is this model?"
"How useful is it?"

0.1 Model Basics

This section will show you some common terms related to statistical models. It may help if you keep the used cars in mind as an example.

What Is a Model?

response

In this book a statistical model is an equation that relates an outcome Y, called the **response**, to one or more other variables X, or X_1, X_2, ...

predictors
model

These "X-variables" are called **predictors**, or **explanatory variables**, or **independent variables**.* The **model** is a mathematical equation

$$DATA = MODEL + ERROR$$
$$Y = f(X) + \epsilon$$

For the used cars, our goal is to predict the price (response) from the mileage (predictor) or perhaps from all three of mileage, age, and make (explanatory variables).

We assume you have already seen examples of models in your beginning statistics course.

- For comparing two groups, the explanatory variable is the group number, $X = 1$ or $X = 2$. The model tells the group mean, either μ_1 for group 1 ($X = 1$) or μ_2 for group 2 ($X = 2$).
- For fitting a line to the points (x, y) of a scatterplot, the model is the equation $Y = \beta_0 + \beta_1 X$, for a line with intercept β_0 and slope β_1.

What is the error term? The ϵ in the model is called "error," not in the sense of mistake, but instead meaning "deviation from the model value." Remember Box: "All models are wrong." The error term tells "How far wrong?" As you may know from introductory statistics, many statistical models assume that the "errors" follow a probability distribution, typically a normal distribution. (More in Chapter 1.)

"Explain" does not mean what you might think. The words of statistics can sometimes mislead. Just as "error" does not mean mistake, the terms *explain* and *account for* do not have their everyday meaning. They are technical terms that refer to "percent reduction in error variation." (If we tell you that for the 50 U.S. states, the variability in median income "explains" 70% of the variability in poverty rate, we haven't really explained anything.)

Underlying models for inference. By "inference" we mean "using the data we *can* see to *infer* facts about numbers or relationships we *can't* see." In statistics†, there are two main kinds of inference:

sample
population
cause and effect

- Inference from a **sample** (the data we see) to a **population** (the larger group we hope our sample represents); and
- Inference about **cause and effect**. This kind of inference is easiest with a *randomized experiment*.

For the used cars, we hope that the sample of cars in our data set is typical of all used cars with similar ages, mileage, and make.

randomized experiment
observational units
experimental units
subjects
observational study

The crucial distinction is whether a research study is a randomized experiment or an observational study. In a **randomized experiment**, the researcher manipulates the explanatory variable by assigning the explanatory group or value to the **observational units**. (These observational units may be called **experimental units** or **subjects** in an experiment.) In an **observational study**, the researchers do not assign the explanatory variable but rather passively observe and record its information. This distinction is important because the type of study determines the scope of conclusion that can be drawn. Randomized experiments allow for drawing *cause-and-effect* conclusions. Observational studies, on the other hand, almost always only allow for

*In economics the response Y is called **endogenous**, and the independent variables X are called **exogenous**.

†Modeling ideas and inference concepts that pertain to statistics also pertain to data science, which sits at the intersection of statistics, computer science, and application.

concluding that variables are *associated*. Ideally, an observational study will anticipate alternative explanations for an association and include the additional relevant variables in the model. These additional explanatory variables are then called **covariates**.

covariates

Regardless of the type of study, the theoretical requirements for inference are necessarily rigorous: samples must be chosen at random from the population; conditions must be assigned at random to subjects or other experimental units. Fortunately, inference is just one use of statistical models. Many uses do not demand so much of your data. Also, particularly with observational studies like the used cars, you may be able to offer nonstatistical reasons to regard the sample *as if* it had been chosen at random.

When we use models for inference, it is important to have different words for two kinds of numbers,

statistic (1) a number we compute from the data is a **statistic** and

parameter (2) the hidden number we really want to know is a **parameter**.

If our goal is to use a sample for inference about a population, for example, the average from the sample is our statistic—we can compute it from the data; the average for the whole population is a parameter—we can't compute it, but its value is what we really want to know.

Statistical Data

case
variable
quantitative
categorical
Cars17

Cases and variables. Table 0.1 shows the first few rows of the data for used cars found in the dataset **ThreeCars2017**. Each row is a **case**, here a single car. Each column is a **variable**: *Price*, *Mileage*, *Age*, and *CarType*. For the car data our goal is to predict price (response) using mileage, age, and model (the explanatory variables). *Price*, *Mileage*, and *Age* are **quantitative**. You can use arithmetic to compare values: "This car has twice the mileage and costs $2500 less than that one." The make of the car is **categorical**. You can classify a car as a Mazda3 or Maxima, but it makes no sense to subtract a Mazda3 from a Maxima. (As an aside, ZIP codes and sports jersey numbers are numerical, but as variables they are categorical, not quantitative. It makes no sense to subtract your ZIP code from mine.)

CarType	Age	Price	Mileage
Mazda3	3	15.9	17.8
Mazda3	2	16.4	19.0
Mazda3	1	18.9	20.9
Mazda3	2	16.9	24.0
Mazda3	2	20.5	24.0
Mazda3	1	19.0	24.2
Mazda3	2	17.5	30.1
Mazda3	3	18.0	32.0
Mazda3	3	13.6	34.8
⋮	⋮	⋮	⋮

TABLE 0.1 The first 9 cases in the **ThreeCars2017** dataset

EXAMPLE 0.1 **Response or explanatory, quantitative or categorical?**

binary

a. Medical school admissions. The response is admission (Yes/No), which is categorical with just two categories, or **binary**. One explanatory variable is GPA, which is quantitative. The other explanatory variable is gender (male/female), which is binary.

b. Ice cream. The response is amount of ice cream taken, which is quantitative. There are two explanatory variables. Bowl size (large or small) is binary. Spoon size (large or small) is also binary.

c. Golf scores. The response is average number of strokes per hole, which is quantitative. There are many possible predictors, depending on how you choose to measure accuracy. Driving distance is clearly quantitative. Putting accuracy might be quantitative (distance from the cup at the end of the putt), but could be binary (yes/no, i.e., make/miss).

d. Migraines. For this example, the explanatory variable is clear: electronic pulses, yes or no. Equally clearly, this variable is binary. The response is not so clear. We want the response to measure relief from pain, but it is far from clear what measure to use.

The pairs of terms (response/predictor, quantitative/categorical) offer a way to classify statistical models into three families. Our three main groups of chapters, Units A, B, and C, are devoted to these families.

Families of Models

This book will show you three main groups of models, plus a bonus:

- Fitting equations to data (**Regression**, Chapters 1–4).
  - The response is quantitative.
  - The predictors are often quantitative, but can also be categorical.
- Comparing groups (**Analysis of Variance**, Chapters 5–8).
  - The response is quantitative.
  - The explanatory variables are almost always categorical. The exception is Analysis of Covariance, which includes both categorical and quantitative predictors.
- Estimating the chances for Yes/No outcomes (**Logistic Regression**, Chapters 9–11).
  - The response must be Yes/No.
  - The predictor(s) can be of either type.

time series

The final bonus Chapter 12, on **time series**, will describe two kinds of models for a response variable that changes over time.

We now turn to an extended example. Our goal is to show you how we rely on a set of four steps to choose, to fit, to assess, and to use a statistical model. We rely on these same four steps throughout the book.

0.2 A Four-Step Process

We will employ a four-step process for statistical modeling throughout this book. These steps are:

- **Choose** a form for the model.

- **Fit** that model to the data.
- **Assess** how well the model fits the data.
- **Use** the model to address the question that motivated the data collection in the first place.

The specific details for how to carry out these steps will differ depending on the type of analysis being performed and, to some extent, on the context of the data being analyzed. But these four steps are carried out in some form in all statistical modeling endeavors. In more detail, the four steps are:

- **Choose** a form for the model. This involves identifying the response and explanatory variable(s) and their types. Typically, we plot the data and explore the patterns we see. A major goal is to describe those patterns and select a model for the relationship between the response and explanatory variable(s).*
- **Fit** that model to the data. This usually entails estimating model parameters based on the sample data. We will almost always use statistical software to do the necessary number-crunching to fit models to data.
- **Assess** how well the model fits the data and meets our needs. How we assess a model depends on how we plan to use it.

  1. For many studies, the main goal of a model is to describe or summarize patterns in the data, and our assessment should focus on how well the model captures those patterns. We can use a variety of **diagnostic plots** to assess

     diagnostic plots

     a. the patterns of **centers** (does our chosen equation track that pattern, or are there features of the data that our model has missed?);

     centers

     b. the pattern of the **spreads** (is the spread in response values roughly uniform across the values of the explanatory variable, so that a single standard deviation is all we need to describe the spread?); and

     spreads

     c. the **shape** of the residuals (i.e., the shape of deviations from the overall pattern).

     shape

     If possible, we look for a scale for which the pattern of centers is simple, the spreads are roughly uniform, residuals follow a normal curve, at least approximately, and our model equation captures all but the normal-shaped residuals.†

  2. For some studies, the goal may be more ambitious: formal (probability-based) inference from the sample data to the larger population or process from which the sample was collected. To justify the use of probability-based methods, there are two additional requirements above and beyond those listed in (1) above.

     a. Randomness. Is the use of a probability model justified?
     b. Independence. Is it reasonable to think of each of the data points as being independent of the other data points?

  We will examine several tools and techniques to use, but the process of assessing model adequacy can be considered as much art as it is science.

- **Use** the model to address the question that motivated collecting the data in the first place. This might be to make predictions, or explain relationships, or assess differences, bearing in mind possible limitations on the scope of

*This might involve transforming one or more of the variables. First choose a suitable scale, then choose an equation. We say more about this in Section 1.4.

†Many inference procedures are based on having errors that are normally distributed, or at least close to normally distributed.

inferences that can be made. For example, if the data were collected as a random sample from a population, then inference can be extended to that population. If treatments were assigned at random to subjects, then a cause-and-effect relationship can be inferred. But if the data arose in other ways, then we have little statistical basis for drawing such conclusions.

To illustrate the process, we consider an example in the familiar setting of a two-sample t-procedure.

EXAMPLE 0.2

WtLoss4

Financial incentives for weight loss Losing weight is an important goal for many individuals. One group of researchers[1] investigated whether financial incentives would help people lose weight more successfully. Some participants in the study were randomly assigned to a treatment group that offered financial incentives for achieving weight-loss goals, while others were assigned to a control group that did not use financial incentives. All participants were monitored over a four-month period, and the net weight change (*Before* − *After*, in pounds) was recorded for each individual. Note that a positive value corresponds to weight loss and a negative value indicates weight gain. The data are given in Table 0.2 and stored in **WeightLossIncentive4**.

Control	12.5	12.0	1.0	−5.0	3.0	−5.0	7.5	−2.5	20.0	−1.0
	2.0	4.5	−2.0	−17.0	19.0	−2.0	12.0	10.5	5.0	
Incentive	25.5	24.0	8.0	15.5	21.0	4.5	30.0	7.5	10.0	18.0
	5.0	−0.5	27.0	6.0	25.5	21.0	18.5			

TABLE 0.2 Weight loss after four months (pounds)

The response variable in this study (weight change) is quantitative and the explanatory variable of interest (control versus incentive) is categorical and binary. The subjects were assigned to the groups at random, so this is a randomized experiment. We may investigate whether there is a statistically significant difference in the distribution of weight changes due to the use of a financial incentive.

CHOOSE

When choosing a form for the model, we consider the question of interest and types of variables involved, then look at graphical displays, and compute summary statistics for the data. Because the weight-loss incentive study has a binary explanatory variable and quantitative response, we examine dotplots and boxplots of the weight losses for each of the two groups (Figure 0.1) and calculate the sample mean and standard deviation for each group.*

Variable	Group	N	Mean	StDev
WeightLoss	Control	19	3.92	9.11
	Incentive	17	15.68	9.41

The dotplots and boxplots show a pair of reasonably symmetric distributions with roughly the same variability in weight loss for the two groups. The mean weight loss for the incentive group is larger than the mean for the control group. One model for these data would be for the weight losses to

*We present vertical dotplots (and boxplots). Horizontal dotplots are common and work perfectly well (and some software packages do not have a simple option for creating vertical dotplots). We prefer vertical dotplots so as to match the vertical boxplots, which foreshadow graphs we will use in Chapter 1.

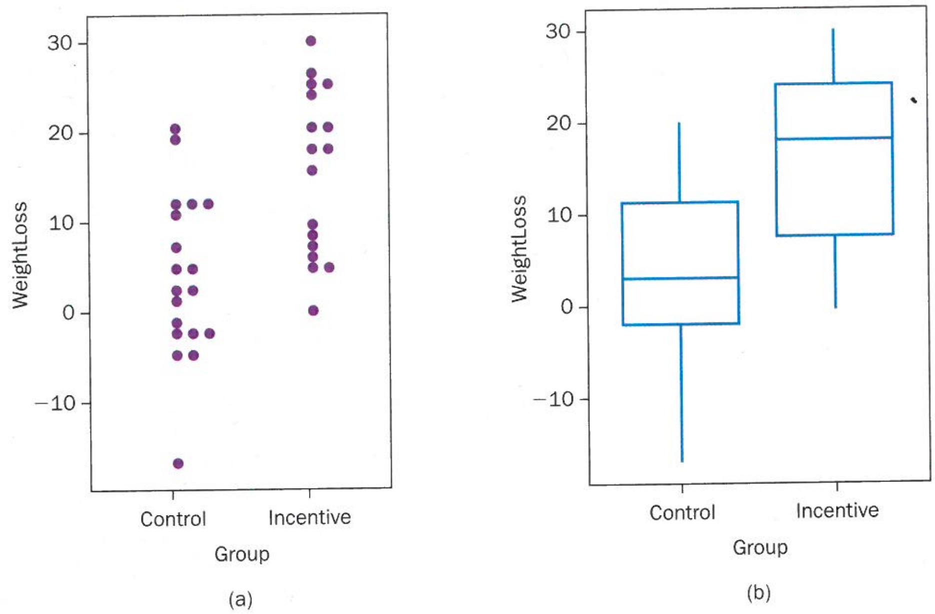

FIGURE 0.1 Weight loss for *Control* versus *Incentive* groups

come from a pair of normal distributions, with different means (and perhaps different standard deviations) for the two groups. Let the parameter μ_1 denote the mean weight loss after four months without a financial incentive, and let μ_2 be the mean with the incentive. If σ_1 and σ_2 are the respective standard deviations and we let the variable Y denote the weight losses, we can summarize the model as $Y \sim N(\mu_i, \sigma_i)$, where the subscript indicates the group and the symbol $\sim$ signifies that the variable has a particular probability distribution, such as normal (denoted in this case with $N(\mu_i, \sigma_i)$).* To see this in the $DATA = MODEL + ERROR$ format, this model could also be written as

$$Y = \mu_i + \epsilon$$

where μ_i is the population mean for the i^{th} group and the random error term $\epsilon \sim N(0, \sigma_i)$. Because we only have two groups, this model says that

$$Y = (\mu_1 + \epsilon) \sim N(\mu_1, \sigma_1) \quad \text{for individuals in the control group}$$
$$Y = (\mu_2 + \epsilon) \sim N(\mu_2, \sigma_2) \quad \text{for individuals in the incentive group}$$

FIT

To fit this model, we need to estimate four parameters (the population means and standard deviations for each of the two groups) using the data from the experiment. The observed means and standard deviations from the two samples provide obvious estimates. We let $\bar{y}_1 = 3.92$ estimate the population mean weight loss for a population getting no incentive (control) and $\bar{y}_2 = 15.68$ estimate the mean for a population getting the incentive. Similarly, $s_1 = 9.11$ and $s_2 = 9.41$ estimate the respective (population) standard deviations. The fitted model (a prediction for the typical weight loss in either group) can then be expressed as

$$\hat{y} = \bar{y}_i$$

that is, that $\hat{y} = 3.92$ pounds for individuals without the incentive and $\hat{y} = 15.68$ pounds for those with the incentive.†

*For this example, an assumption that the variances are equal, $\sigma_1^2 = \sigma_2^2$, might be reasonable, but that would lead to the less familiar pooled variance version of the t-test.

†We use the carat symbol ^ above a variable name to indicate predicted value, and refer to this as "y-hat."

Note that the error term does not appear in the fitted model since, when predicting a particular weight loss, we don't know whether the random error will be positive or negative. That does not mean that we expect there to be no error, just that the best guess for the *average* weight loss under either condition is the sample group mean, $\bar{y}_i$.

ASSESS

Our goal is to test a particular scientific hypothesis—that the financial incentive leads to greater weight loss—and to estimate the size of the effect of the incentive. The model we have presented expects that departures from the mean in each group (the random errors) should follow a normal distribution with mean zero. To assess this, we examine the sample *residuals* (deviations between the actual data weight losses and those predicted by the model):

$$\textit{residual} = \text{observed} - \text{predicted} = y - \hat{y}$$

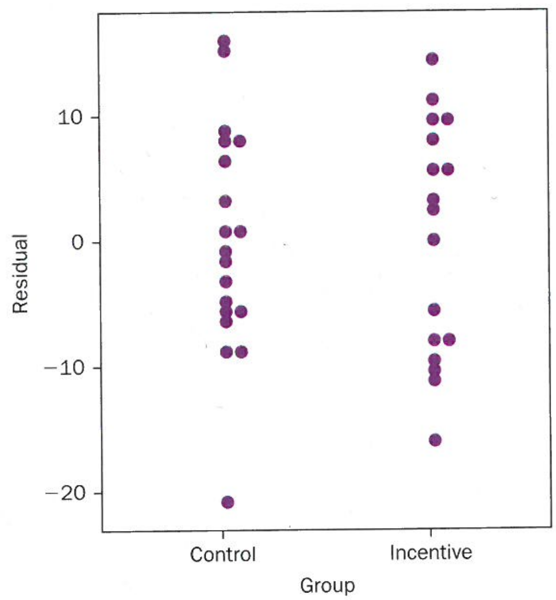

FIGURE 0.2 Residuals from group weight-loss means

The residuals are calculated by subtracting $\hat{y} = 3.92$ from each individual's weight loss in the control group and $\hat{y} = 15.68$ for each individual in the incentive group. Dotplots of the residuals for each group are shown in Figure 0.2. Note that the distributions of the residuals are the same as the original weight-loss data, except that both are shifted to have a mean of zero. We don't see any significant departures from normality in the dotplots, but it's difficult to judge normality from dotplots with so few observations. Normal quantile plots (as shown in Figure 0.3) are a more informative technique for assessing normality. Departures from a linear trend in such plots indicate a lack of normality. Figure 0.3 shows no substantial departure. Normal quantile plots will be examined in more detail in the next chapter.

As a second component of assessment, we consider whether a simpler model might fit the data essentially as well as our model with different means for each group. This is analogous to testing the standard hypotheses for a two-sample t-test:

$H_0 : \mu_1 = \mu_2$
$H_a : \mu_1 \neq \mu_2$

The null hypothesis (H_0) corresponds to the simpler model $Y = \mu + \epsilon$, which uses the same mean for both the control and incentive groups. The

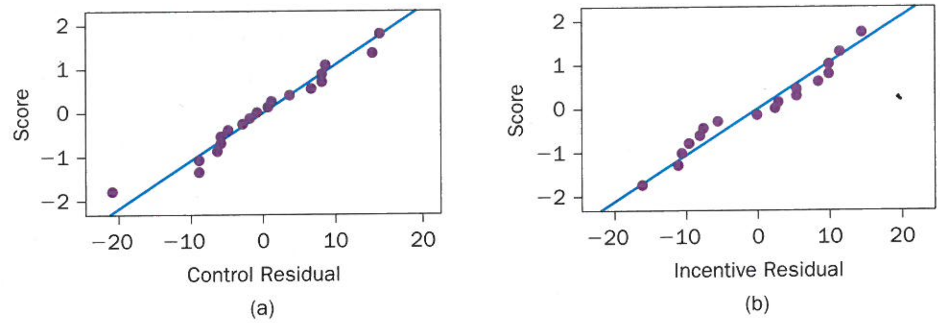

FIGURE 0.3 Normal quantile plots for residuals of weight loss

alternative (H_a) reflects the model that allows each group to have a different mean. Would the simpler (common mean) model suffice for the weight-loss data, or do the two separate group means provide a better explanation for the data? One way to judge this is with the usual two-sample t-test as shown in the computer output below.

```
Two-sample T for WeightLoss

Group       N    Mean   StDev  SE Mean
Control    19    3.92    9.11      2.1
Incentive  17   15.68    9.41      2.3

Difference = mu (Control) - mu (Incentive)
Estimate for difference: -11.76
95% CI for difference: (-18.05, -5.46)
T-Test of difference = 0 (vs not =): T-Value = -3.80 P-Value = 0.001 DF = 33
```

The extreme value for this test statistic ($t = -3.80$) produces a very small P-value (0.001). If financial incentive has no effect on how much weight people lose, we would get a value of t as large as 3.80 (in magnitude) about one time out of one thousand. The evidence that financial incentive makes a difference (increasing mean weight loss) is strong.* We prefer the statistical model that allows for different group means, despite its additional complexity, over the model that uses the same mean for both groups.†

Earlier (when first discussing the ASSESS step) we said that the formal (probability-based) inference requires the added conditions of randomness and independence. For the weight-loss data, the participants were randomly assigned to groups and each participant contributed exactly one data point, so the randomness and independence conditions are met.

USE

Because this was a controlled experiment with random assignment of the control and incentive conditions to the subjects, and because the data produced a very small P-value, we can infer that the financial incentives did produce a difference in the average weight loss over the four-month period; that is, the random allocation of conditions to subjects allows us to draw a cause-and-effect relationship.

This test based on a P-value addresses just one of three kinds of questions we often ask to summarize a fitted model.

*See Section 2.1 for a more complete discussion of interpreting P-values.

†A note about terminology: When the P-value is small, we say that there is "statistically significant" evidence that the population means are different. We might say "the *sample* means are (statistically) significantly different." That is *not* the same as saying that any difference between the two *population* means has practical importance.

Tests, Intervals, and Effect Sizes

- **Tests:** Is there an effect?
  Does the incentive have an effect? Does our model need separate means for the incentive and control groups?
- **Intervals:** How big is the effect?
  How precisely can we estimate the size of the effect? What is our margin of error?
- **Effect sizes:** Is the effect big enough to matter?

Tests: As you have seen, a t-test and P-value tell us yes, there is strong evidence of an effect due to the incentive. We *do* need two separate means in our model. *But—the* P*-value alone tells only the strength of the evidence that the means differ. The* P*-value tells nothing about the size of the difference between the two groups.*

Intervals: The difference in group means (incentive minus control) is 11.76 pounds. As you may remember (but we will show you later, in case you don't remember), there is a way to use the data to compute a margin of error, which here turns out to be 6.3. We use the margin of error to construct a 95% confidence interval, 11.76 ± 6.3. We can be 95% confident that the incentive is worth between 5.5 and 18.1 pounds of additional weight loss over four months on average. The interval tells you about precision, but not about importance.

Effect sizes: At this point we know from the P-value that the difference of 11.76 pounds is too big to be due just to the random assignment of subjects to groups. We know from the margin of error that with 95% confidence, the incentive is worth between 5.5 and 18.1 pounds. What we don't yet know is how much a difference of 11.76 pounds actually matters.

To address this question, we need a yardstick, a standard of comparison for judging the importance of 11.76. The usual choice is to use the typical size of person-to-person variation. We hope "typical size" and "variation" bring to mind "standard deviation." The only question left is "Which standard deviation?" There are several choices, and several definitions of effect size, but for now we just note the disagreement and leave it to others to argue about whose weeds matter and whose weeds don't.* To keep it simple, we use the pooled standard deviation of 9.25 as our yardstick.

Our effect size is the ratio of the observed difference (11.76) to the standard deviation (9.25) for person-to-person differences. This ratio, $11.76/9.25 = 1.27$, tells us that the effect of the incentive is 127% as big as the typical size of the difference between you and an average person.

$$\text{effect size} = \frac{|\bar{y}_1 - \bar{y}_2|}{s_p} = \frac{|15.68 - 3.92|}{9.25} = 1.27$$

Such a difference, 127%, is quite large. (General guidelines are that an effect size of 0.2 or 20% is small, an effect size of 0.5 or 50% is medium, and an effect size of 1.0 or 100% is large.)

Before leaving this example, we note three cautions:

- First, all but two of the participants in this study were adult men, so we should avoid drawing conclusions about the effect of financial incentives on weight loss in women.

*There are many ways to calculate effect size $= \frac{|\bar{y}_1 - \bar{y}_2|}{s}$. Using the pooled standard deviation $s_p = \sqrt{\frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}}$ gives a number that is called Hedge's g (which is very closely related to a measure called Cohen's d). Some people prefer to use $s_{Control}$ in place of s_p; this is called Glass' Δ (delta). Others recommend using the larger of the two sample s_i's.

- Second, if the participants in the study were not selected by taking a random sample, we would have difficulty justifying a statistical basis for generalizing the findings to other adults. Any such generalization must be justified on other grounds (such as a belief that most adults respond to financial incentives in similar ways).
- Third, the experimenters followed up with subjects to see if weight losses were maintained seven months after the start of the study (and three months after any incentives expired).

The results from the follow-up study appear in Exercise 0.23.

CHAPTER SUMMARY

In this chapter, we reviewed basic terminology, introduced the four-step approach to modeling that will be used throughout the text, and revisited a common two-sample inference problem.

After completing this chapter, you should be able to distinguish between a **sample** and a **population**, describe the difference between a **parameter** and a **statistic**, and identify variables as **categorical** or **quantitative**. Prediction is a major component to modeling, so identifying **explanatory** (or predictor) **variables** that can be used to develop a model for the **response variable** is an important skill. Another important idea is the distinction between **observational studies** (where researchers simply observe what is happening) and **experiments** (where researchers impose "treatments").

We introduced the fundamental idea that a **statistical model** partitions data into two components, one for the model and one for error. Even though the models will get more complex as we move through the more advanced settings, this statistical modeling idea will be a recurring theme throughout the text. The error term and conditions associated with this term are important features in distinguishing statistical models from mathematical models. You saw how to compute **residuals** by comparing the observed data to predictions from a model as a way to begin quantifying the errors.

The **four-step process** of choosing, fitting, assessing, and using a model is vital. Each step in the process requires careful thought, and the computations will often be the easiest part of the entire process. Identifying the response and explanatory variable(s) and their types (categorical or quantitative) helps us **choose** the appropriate model(s). Statistical software will almost always be used to **fit** models and obtain estimates. Comparing models and **assessing** the adequacy of these models will require a considerable amount of practice, and this is a skill that you will develop over time. Try to remember that **using** the model to make predictions, explain relationships, or assess differences is only one part of the four-step process.

EXERCISES

Conceptual Exercises

0.1 Categorical or quantitative? Suppose that a statistics professor records the following for each student enrolled in her class:

- Gender
- Major
- Score on first exam
- Number of quizzes taken (a measure of class attendance)
- Time spent sleeping the previous night
- Handedness (left- or right-handed)
- Political inclination (liberal, moderate, or conservative)
- Time spent on the final exam
- Score on the final exam

For the following questions, identify the response variable and the explanatory variable(s). Also classify each variable as quantitative or categorical. For categorical variables, also indicate whether the variable is binary.

a. Do the various majors differ with regard to average sleeping time?

b. Is a student's score on the first exam useful for predicting his or her score on the final exam?

c. Do male and female students differ with regard to the average time they spend on the final exam?

d. Can we tell much about a student's handedness by knowing his or her major, gender, and time spent on the final exam?

0.2 More categorical or quantitative? Refer to the data described in Exercise 0.1 that a statistics professor records for her students. For the following questions, identify the response variable and the explanatory variable(s). Also, classify each variable as quantitative or categorical. For categorical variables, also indicate whether the variable is binary.

a. Do the proportions of left-handers differ between males and females on campus?

b. Are sleeping time, exam 1 score, and number of quizzes taken useful for predicting time spent on the final exam?

c. Does knowing a student's gender help predict his or her major?

d. Does knowing a student's political inclination and time spent sleeping help predict his or her gender?

0.3 Sports projects. For each of the following sports-related projects, identify observational units and the response and explanatory variables when appropriate. Also, classify the variables as quantitative or categorical.

a. Interested in predicting how long it takes to play a Major League Baseball game, an individual recorded the following information for all 14 games played on August 11, 2017: time to complete the game, total number of runs scored, margin of victory, total number of pitchers used, ballpark attendance at the game, and which league (National or American) the teams were in.

b. Over the course of several years, a golfer kept track of the length of all of his putts and whether or not he made the putt. He was interested in predicting whether or not he would make a putt based on how long it was.

c. Some students recorded lots of information about all of the football games played by Drew Brees during the 2016 season. They recorded his passing yards, number of pass attempts, number of completions, and number of touchdown passes.

0.4 More sports projects. For each of the following sports-related projects, identify observational units and the response and explanatory variables when appropriate. Also, classify the variables as quantitative or categorical.

a. A volleyball coach wants to see if a player using a jump serve is more likely to lead to winning a point than using a standard overhand serve.

b. To investigate whether the "home-field advantage" differs across major team sports, researchers kept track of how often the home team won a game for all games played in the 2016 and 2017 seasons in Major League Baseball, National Football League, National Basketball Association, and National Hockey League.

c. A student compared men and women professional golfers on how far they drive a golf ball (on average) and the percentage of their drives that hit the fairway.

0.5 Scooping ice cream. In a study reported in the *American Journal of Preventive Medicine*[2] 85 nutrition experts were asked to scoop themselves as much ice cream as they wanted. Some of them were randomly given a large bowl (34 ounces) as they entered the line, and the others were given a smaller bowl (17 ounces). Similarly, some were randomly given a large spoon (3 ounces) and the others were given a small spoon (2 ounces). Researchers then recorded how much ice cream each subject scooped for him- or herself. Their conjecture was that those given a larger bowl would tend to scoop more ice cream, as would those given a larger spoon.

a. Identify the observational units in this study.

b. Is this an observational study or a controlled experiment? Explain how you know.

c. Identify the response variable in this study, and classify it as quantitative or categorical.

d. Identify the explanatory variable(s) in this study, and classify it(them) as quantitative or categorical.

0.6 Diet plans. An article in the *Journal of the American Medical Association*[3] reported on a study in which 160 subjects were randomly assigned to one of four popular diet plans: Atkins, Ornish, Weight Watchers, and Zone. Among the variables measured were:

- Which diet the subject was assigned to
- Whether or not the subject completed the 12-month study
- The subject's weight loss after 2 months, 6 months, and 12 months (in kilograms, with a negative value indicating weight gain)

a. Classify each of these variables as quantitative or categorical.

b. The primary goal of the study was to investigate whether weight loss tends to differ significantly among the four diets. Identify the explanatory and response variables for investigating this question.

c. Is this an observational study or a controlled experiment? Explain how you know.

0.7 Wine model. In his book *SuperCrunchers: Why Thinking by Numbers Is the New Way to Be Smart*, Ian Ayres wrote about Orley Ashenfelter, who gained fame and generated considerable controversy by using statistical models to predict the quality of wine.

Ashenfelter developed a model based on decades of data from France's Bordeaux region, which Ayres reports as

$$\begin{aligned} WineQuality = 12.145 &+ 0.00117 WinterRain \\ &+ 0.0614 AverageTemp \\ &- 0.00386 HarvestRain + \epsilon \end{aligned}$$

where *WineQuality* is a function of the price, rainfall is measured in millimeters, and temperature is measured in degrees Celsius.

a. Identify the response variable in this model. Is it quantitative or categorical?

b. Identify the explanatory variables in this model. Are they quantitative or categorical?

c. According to this model, is higher wine quality associated with more or with less winter rainfall?

d. According to this model, is higher wine quality associated with more or with less harvest rainfall?

e. According to this model, is higher wine quality associated with more or with less average growing season temperature?

f. Are the data that Ashenfelter analyzed observational or experimental? Explain.

0.8 Predicting NFL wins. Consider the following model for predicting the number of games that a National Football League (NFL) team wins in a season:

$$Wins = 3.6 + 0.5PF - 0.3PA + \epsilon$$

where *PF* stands for average points a team scores per game over an entire season and *PA* stands for points allowed per game. Each team plays 16 games in a season.

a. Identify the response variable in this model. Is it quantitative or categorical?

b. Identify the explanatory variables in this model. Are they quantitative or categorical?

c. According to this model, how many more wins is a team expected to achieve in a season if they increase their scoring by an average of 3 points per game?

d. According to this model, how many more wins is a team expected to achieve in a season if they decrease their points allowed by an average of 3 points per game?

e. Based on your answers to parts (c) and (d), does it seem that a team should focus more on improving its offense or improving its defense?

f. Are the data analyzed for this study observational or experimental?

0.9 Measuring students. The registrar at a small liberal arts college computes descriptive summaries for all members of the entering class on a regular basis. For example, the mean and standard deviation of the high school grade point averages for all entering students in a particular year were 3.16 and 0.5247, respectively. The Mathematics Department is interested in helping all students who want to take mathematics to identify the appropriate course, so they offer a placement exam. A randomly selected subset of students taking this exam during the past decade had an average score of 71.05 with a standard deviation of 8.96.

a. What is the population of interest to the registrar at this college?

b. Are the descriptive summaries computed by the registrar (3.16 and 0.5247) statistics or parameters? Explain.

c. What is the population of interest to the Mathematics Department?

d. Are the numerical summaries (71.05 and 8.96) statistics or parameters? Explain.

0.10 Measuring pumpkins. A pumpkin farmer knows that to sell his pumpkins to the local grocery store, each pumpkin he sells must weigh at least 2 pounds. He keeps track of the proportion of pumpkins he is able to sell to the store and at the end of the season finds that 91% of his pumpkins met the 2-pound cutoff. A customer at the store is more interested in the number of pumpkin seeds inside the pumpkins at the store. She purchases 20 pumpkins over the season and carefully counts how many seeds are in each pumpkin. The mean number of seeds she records is 123.2.

a. What is the population of interest to the farmer?

b. Is the descriptive summary computed by the farmer (91%) a statistic or a parameter? Explain.

c. What is the population of interest to the customer?

d. Is the numerical summary (123.2) a statistic or a parameter? Explain.

Guided Exercises

0.11 Scooping ice cream. Refer to Exercise 0.5 on self-serving ice cream. The following table reports the average amounts of ice cream scooped (in ounces) for the various treatments:

	17-ounce bowl	34-ounce bowl
2-ounce spoon	4.38	5.07
3-ounce spoon	5.81	6.58

a. Does it appear that the size of the bowl had an effect on amount scooped? Explain.

b. Does it appear that the size of the spoon had an effect on amount scooped? Explain.

c. Which appears to have more of an effect: size of bowl or size of spoon? Explain.

d. Does it appear that the effect of the bowl size is similar for both spoon sizes, or does it appear that the effect of the bowl size differs substantially for the two spoon sizes? Explain.

0.12 Diet plans. The study discussed in Exercise 0.6 included another variable: the degree to which the subject adhered to the assigned diet, taken as the average of 12 monthly ratings, each on a 1–10 scale (with 1 indicating complete nonadherence and 10 indicating full adherence). As discussed in Exercise 0.6 the primary goal of the study was to investigate whether weight loss tends to differ significantly among the four diets. A secondary goal was to investigate whether weight loss is affected by the adherence level.

a. Identify the explanatory and response variables for investigating the secondary question.

b. If the researchers' analysis of the data leads them to conclude that there is a significant difference in weight loss among the four diets, can they legitimately conclude that the difference is because of the diet? Explain why or why not.

c. If the researchers' analysis of the data analysis leads them to conclude that there is a significant association between weight loss and adherence level, can they legitimately conclude that a cause-and-effect association exists between them? Explain why or why not.

0.13 Predicting NFL wins: Packers/Giants. Consider again, the model for predicting the number of games that a National Football League (NFL) team wins in a season:

$$Wins = 3.6 + 0.5PF - 0.3PA + \epsilon$$

where PF stands for average points a team scores per game over an entire season and PA stands for points allowed per game. Each team plays 16 games in a season.

a. The Green Bay Packers had a good regular season record in 2016, winning 10 games and losing 6. They averaged 27.0 points scored per game, while allowing an average of 24.25 points per game against them. How many wins does this model predict the Green Bay Packers had in 2016?

b. Find the residual for the Green Bay Packers in 2016 using this model.

c. The largest positive residual value from this model for the 2016 season belongs to the New York Giants, with a residual value of 3.04 games. The Giants actually won 11 games. Determine this model's predicted number of wins for the Giants.

0.14 Predicting NFL wins: Patriots/Chargers. Refer to the model in Exercise 0.13 for predicting the number of games won in a 16-game NFL season based on the average number of points scored per game (PF) and average number of points allowed per game (PA).

a. Use the model to predict the number of wins for the 2016 New England Patriots, who scored 441 points and allowed 250 points in their 16 games.

b. The Patriots actually won 14 games in 2016. Determine their residual from this model, and interpret what this means.

c. The largest negative residual value from this model for the 2016 season belongs to the San Diego Chargers, with a residual value of −3.48 games. Interpret what this residual means.

0.15 Roller coasters. The Roller Coaster Database (rcdb.com) contains lots of information about roller coasters all over the world. The following statistical model for predicting the top speed (in miles per hour) of a coaster was based on more than 100 roller coasters in the United States and data displayed on the database:

$$TopSpeed = 54 + 7.6TypeCode + \epsilon$$

where $TypeCode = 1$ for steel roller coasters and $TypeCode = 0$ for wooden roller coasters.

a. What top speed does this model predict for a wooden roller coaster?

b. What top speed does this model predict for a steel roller coaster?

c. Determine the difference in predicted speeds in miles per hour for the two types of coasters. Also identify where this number appears in the model equation, and explain why that makes sense.

0.16 Roller coasters: multiple predictors. Refer to the information about roller coasters in Exercise 0.15. Some other predictor variables available at the database include: age, total length, maximum height, and maximum vertical drop. Suppose that we include all of these predictor variables in a statistical model for predicting the top speed of the coaster.

a. For each of these predictor variables, indicate whether you expect its coefficient to be positive or negative. Explain your reasoning for each variable.

b. Which of these predictor variables do you expect to be the best single variable for predicting a roller coaster's top speed? Explain why you think that.

The following statistical model was produced from these data:

$$\begin{aligned}Speed = 33.4 &+ 0.10Height + 0.11Drop \\ &+ 0.0007Length - 0.023Age - 2.0TypeCode + \epsilon\end{aligned}$$

c. Comment on whether the signs of the coefficients are as you expect.

d. What top speed would this model predict for a steel roller coaster that is 10 years old, with a maximum height of 150 feet, maximum vertical drop of 100 feet, and length of 4000 feet?

Handwriting and gender. The file **Handwriting** contains survey data from a sample of 203 statistics students at Clarke University. Each student was given 25 handwriting specimens and they were to guess, as best they could, the gender (male or female) of the person who penned the specimen. (Each specimen was a handwritten address, as might appear on the envelope of a letter. There were no semantic clues in the specimen that could tip off the author's gender.) The survey was repeated two times—the second time was the same 25 specimens given in a randomly new order. Exercises 0.17 to 0.22 use the data from this study. **Handwrt**

0.17 CHOOSE/FIT: Survey1. One variable given in the **Handwriting** dataset is *Survey1*, which records the percent correct that each individual had on the first survey.

a. CHOOSE: Construct a histogram of the variable *Survey1*, and discuss what it tells you about the percent correct on the first survey.

b. CHOOSE: Do you think that the subjects are better in guessing the author's gender than mere 50-50 coin flipping would be? Write out the model that describes this situation.

c. FIT: Compute the summary statistics for the sample and give the fitted model.

0.18 CHOOSE/FIT: Survey2. One variable given in the **Handwriting** dataset is *Survey2*, which records the percent correct that each individual had on the second survey.

a. CHOOSE: Construct a histogram of the variable *Survey2*, and discuss what it tells you about the percent correct on the second survey.

b. CHOOSE: Do you think that the subjects' guesses are better in guessing the author's gender than mere 50-50 coin flipping would be? Write out the model that describes this situation.

c. FIT: Compute the summary statistics for the sample and give the fitted model.

0.19 ASSESS/USE: Survey1. Return to the variable *Survey1*.

a. ASSESS: Construct an appropriate plot of the residuals, and comment on whether the graph supports the idea that the residuals are normally distributed.

b. ASSESS: Do you think that the subjects are better in guessing the author's gender than mere 50-50 coin flipping would be? Assess whether our model is better than a coin-flipping model by performing a one-sample *t*-test.

c. USE: Give the conclusion in context to the test performed in part (b).

0.20 ASSESS/USE: Survey2. Return to the variable *Survey2*.

a. ASSESS: Construct an appropriate plot of the residuals and comment on whether the graph supports the idea that the residuals are normally distributed.

b. ASSESS: Do you think that the subjects are better in guessing the author's gender than mere 50-50 coin flipping would be? Assess whether our model is better than a coin-flipping model by performing a one-sample *t*-test.

c. USE: Give the conclusion in context to the test performed in part (b).

0.21 Two-sample Four-step: Survey1. Suppose we want to compare the ability of men and women students in guessing the gender identities of authors of handwriting samples. Consider again the measurements in *Survey1*.

a. CHOOSE: Construct an appropriate plot to compare the guessing abilities of men and women students, discuss what the plot tells you, and write out the model that allows for differences in guessing ability between the two groups.

b. FIT: Compute the appropriate summary statistics and give the fitted model.

c. ASSESS: Construct an appropriate plot of the residuals, and comment on whether the graph supports the idea that the residuals are normally distributed. Also, perform the appropriate two-sample *t*-test to compare the chosen model to the model of no difference between the two groups.

d. USE: Give your conclusion detailing which model you would use and why.

0.22 Two-sample Four-step: Survey2. Suppose we want to compare the ability of men and women students in guessing the gender identities of authors of handwriting samples. Consider again the measurements in *Survey2*.

a. CHOOSE: Construct an appropriate plot to visualize the differences between guessing abilities of men and women students, discuss what the plot tells you, and write out the model that allows for differences in guessing ability between the two groups.

b. FIT: Compute the appropriate summary statistics and give the fitted model.

c. ASSESS: Construct an appropriate plot of the residuals and comment on whether the graph supports the idea that the residuals are normally distributed. Also, perform the appropriate two-sample *t*-test to compare the chosen model to the model of no difference between the two groups.

d. USE: Give your conclusion detailing which model you would use and why.

Open-ended Exercises

0.23 Incentive for weight loss. The study (Volpp et al., 2008) on financial incentives for weight loss in Example 0.2 on page 7 used a follow-up weight check after seven months to see whether weight losses persisted after the original four months of treatment. The results are given in Table 0.3 and in the variable *Month7Loss* of the **WeightLossIncentive7** data file. Note that a few

Table 0.3 Weight loss after seven months (pounds)

Control	−2.0	7.0	19.5	−0.5	−1.5	−10.0	0.5	5.0	8.5	$\bar{y}_1 = 4.64$
	18.0	16.0	−9.0	4.5	23.5	5.5	6.5	−9.5	1.5	$s_1 = 9.84$
Incentive	11.5	20.0	−22.0	2.0	7.5	16.5	19.0	18.0	−1.0	$\bar{y}_2 = 7.80$
	5.5	24.5	9.5	10.0	−8.5	4.5				$s_2 = 12.06$

participants dropped out and were not reweighed at the seven-month point. As with the earlier example, the data are the change in weight (in pounds) from the beginning of the study and positive values correspond to weight losses. Using Example 0.2 as an outline, follow the four-step process to see whether the data provide evidence that the beneficial effects of the financial incentives still apply to the weight losses at the seven-month point. **WtLoss7**

0.24 Statistics student survey: resting pulse rate. An instructor at a small liberal arts college distributed a data collection card, similar to what is shown below, on the first day of class. The data for two different sections of the course are shown in the file **Day1Survey**. Note that the names have not been entered into the dataset. **Day1**

Data Collection Card

Directions: Please answer each question and return to me.

1. Your name (as you prefer): ____________
2. What is your current class standing? ____________
3. Sex: Male ________ Female ________
4. How many miles (approximately) did you travel to get to campus? ____________
5. Height (estimated) in inches: ____________
6. Handedness (Left, Right, Ambidextrous): ____________
7. How much money, in coins (not bills), do you have with you? $ ____________
8. Estimate the length of the white string (in inches): ___
9. Estimate the length of the black string (in inches): ___
10. How much do you expect to read this semester (in pages/week)? ____________
11. How many hours do you watch TV in a typical week? ____________
12. What is your resting pulse? ____________
13. How many text messages have you sent and received in the last 24 hours? ____________

a. Apply the four-step process to the survey data to address the question: "Is there evidence that the mean resting pulse rate for women is different from the mean resting pulse rate for men?"

b. Pick another question that interests you from the survey and compare the responses of men and women.

0.25 Statistics student survey: reading amount. Refer to the survey of statistics students described in Exercise 0.24 with data in **Day1Survey**. Use the survey data to address the question: "Do women expect to do more reading than men?" **Day1**

0.26 Marathon training: short versus long run. Training records for a marathon runner are provided in the file **Marathon**. The *Date*, *Miles* run, *Time* (in minutes:seconds:hundredths), and running *Pace* (in minutes:seconds:hundredths per mile) are given for a five-year period. The time and pace have been converted to decimal minutes in *TimeMin* and *PaceMin*, respectively. Use the four-step process to investigate if a runner has a tendency to go faster on short runs (5 or less miles) than long runs. The variable *Short* in the dataset is coded with 1 for short runs and 0 for longer runs. Assume that the data for this runner can be viewed as a sample for runners of a similar age and ability level. **Marath**

0.27 Marathon training: younger versus older runner. Refer to the data described in Exercise 0.26 that contains five years' worth of daily training information for a runner. One might expect that the running patterns change as the runner gets older. The file **Marathon** also contains a variable called *After*2004, which has the value 0 for any runs during the years 2002–2004 and 1 for runs during 2005 and 2006.* Use the four-step process to see if there is evidence of a difference between these two time periods in the following aspects of the training runs: **Marath**

a. The average running pace (*PaceMin*)

b. The average distance run per day (*Miles*)

0.28 Handwriting. Return to the data file **Handwriting** and suppose we decide upon an operational definition of ability to identify author gender that requires the subject make a correct identification on both surveys. Using the *Both* variable, compare the data to the coin-flipping model using the four-step procedure. **Handwrt**

0.29 Handwriting: males versus females. Consider the **Handwriting** dataset one last time. Suppose we want to compare the ability of men and women students in guessing the gender identities of authors of handwriting samples using their responses to both surveys. Using the *Both* variable, compare the data for men and women students using the four-step procedure. **Handwrt**

0.30 Student survey and marathon pace: effect sizes. Compute and interpret the effect size in each of the following situations.

a. Compare mean resting pulse rate for men and women students, as in Exercise 0.24, for the data in **Day1Survey**. **Day1**

b. Compare mean running pace for long and short training runs, as in Exercise 0.27, for the data in **Marathon**. **Marath**

0.31 Marathon training and handwriting: effect sizes. Compute and interpret the effect size in each of the following situations.

a. Compare mean miles per day for training before and after 2004, as in Exercise 0.27, for the data in **Marathon**. **Marath**

b. Compare mean guessing ability for men and women, as in Exercise 0.29, for the data in **Handwriting**. **Handwrt**

Supplementary Exercises

0.32 Pythagorean theorem of baseball. Renowned baseball statistician Bill James devised a model for predicting a team's winning percentage. Dubbed the "Pythagorean Theorem of Baseball," this model predicts a team's winning percentage as

*These data were collected by one of the authors, when he was training for marathons. Sadly, the data are old because he is old.

$$\text{Winning percentage} = \frac{(\text{runs scored})^2}{(\text{runs scored})^2 + (\text{runs against})^2} \times 100 + \epsilon$$

a. Use this model to predict the winning percentage for the Chicago Cubs, who scored 808 runs and allowed 556 runs in the 2016 season.

b. The Chicago Cubs actually won 103 games and lost 59 in the 2016 season. Determine the winning percentage and also determine the residual from the Pythagorean model (by taking the observed winning percentage minus the predicted winning percentage).

c. Interpret what this residual value means for the 2016 Cubs. (*Hints*: Did the team do better or worse than expected, given their runs scored and runs allowed? By how much?)

d. Repeat parts (a–c) for the 2016 San Diego Padres, who scored 686 runs and allowed 770 runs, while winning 68 games and losing 94 games.

e. Which team (Cubs or Padres) fell short of their Pythagorean expectations by more? Table 0.4 provides data,[4] predictions, and residuals for all 30 Major League Baseball teams in 2016.

f. Which team exceeded their Pythagorean expectations the most? Describe how this team's winning percentage compares to what is predicted by their runs scored and runs allowed.

g. Which team fell furthest below their Pythagorean expectations? Describe how this team's winning percentage compares to what is predicted by their runs scored and runs allowed.

Table 0.4 Winning percentage and Pythagorean predictions for baseball teams in 2016

TEAM	W	L	WinPct	RunScored	RunsAgainst	Predicted	Residual
Arizona Diamondbacks	69	93	42.59	752	890	41.65	0.94
Atlanta Braves	68	93	42.24	649	779	40.97	1.26
Baltimore Orioles	89	73	54.94	744	715	51.99	2.95
Boston Red Sox	93	69	57.41	878	694	61.55	−4.14
Chicago Cubs	103	58	63.98	808	556		
Chicago White Sox	78	84	48.15	686	715	47.93	0.22
Cincinnati Reds	68	94	41.98	716	854	41.28	0.70
Cleveland Indians	94	67	58.39	777	676	56.92	1.47
Colorado Rockies	75	87	46.30	845	860	49.12	−2.82
Detriot Tigers	86	75	53.42	750	721	51.97	1.45
Houston Astros	84	78	51.85	724	701	51.61	0.24
Kansas City Royals	81	81	50.00	675	712	47.33	2.67
Los Angeles Angels	74	88	45.68	717	727	49.31	−3.63
Los Angeles Dodgers	91	71	56.17	725	638	56.36	−0.18
Miami Marlins	79	82	49.07	655	682	47.98	1.09
Milwaukee Brewers	73	89	45.06	671	733	45.59	−0.53
Minnesota Twins	59	103	36.42	722	889	39.74	−3.32
New York Mets	87	75	53.70	671	617	54.19	−0.48
New York Yankees	84	78	51.85	680	702	48.41	3.44
Oakland Athletics	69	93	42.59	653	761	42.41	0.19
Philadelphia Phillies	71	91	43.83	610	796	37.00	6.83
Pittsburgh Pirates	78	83	48.45	729	758	48.05	0.40
San Diego Padres	68	94	41.98	686	770		
Seattle Mariners	86	76	53.09	768	707	54.13	−1.04
San Francisco Giants	87	75	53.70	715	631	56.22	−2.51
St. Louis Cardinals	86	76	53.09	779	712	54.48	−1.40
Tampa Bay Rays	68	94	41.98	672	713	47.04	−5.07
Texas Rangers	95	67	58.64	765	757	50.53	8.12
Toronto Blue Jays	89	73	54.94	759	666	56.50	−1.56
Washington Nationals	95	67	58.64	763	612	60.85	−2.21